NATIONAL TEACHERS COLLEGE
School of Arts, Sciences and Technology
Bachelor of Science in Information Technology

# StudentShield: Machine Learning-Based Emotional Distress Detection and Bullying Reporting System with Automated Parental Alerts for Junior High School Students at Saint Vincent Blessed School of Manila

A Capstone Project
presented to the Faculty of the
School of Arts, Science and Technology
National Teachers College
Quiapo, Manila

In Partial Fulfillment of the
Requirements for the Degree of
Bachelor of Science in Information Technology

By
Enriquez, Justin Altamia
Melgarejo, John Carlo
Meneses, Vonn Christian Miel
Ramos, John Cris
Urayanza, Ashley Princess

November 2026

**APPROVAL SHEET FOR TITLE DEFENSE**

This Capstone Project, entitled StudentShield: Machine Learning-Based Emotional Distress Detection and Bullying Reporting System with Automated Parental Alerts for Junior High School Students at Saint Vincent Blessed School of Manila, was prepared and submitted by:

Enriquez, Justin Altamia
Melgarejo, John Carlo
Meneses, Vonn Christian Miel
Ramos, John Cris
Urayanza, Ashley Princess

has been approved by the REVIEW COMMITTEE as partial fulfillment of the requirements for the subject Capstone Project

Mr. Edward James V. Grageda
Adviser

REVIEW COMMITTEE

Mrs. Rodessa Bunag
Panelist 1

Mr. Joshua Pacer
Panelist 2

Accepted and approved by:

Carlito L. Cunanan, MIT
Information Technology Department

Liezel L. Aguilar, MACDDS
School of Arts, Sciences and Technology

**APPROVAL SHEET FOR PRE - ORAL DEFENSE**

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---

REVIEW COMMITTEE

Mr. Joemel Atiga
Panelist 1

Mr. Marvic B. Tifora
Panelist 2

Mr. Jayvee L. Capiral
Panelist Chair

---

Accepted and approved by:

Carlito L. Cunanan, MIT
Information Technology Department

Liezel L. Aguilar, MACDDS
School of Arts, Sciences and Technology

## ABSTRACT

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# CHAPTER 1

# INTRODUCTION

## Project Context

Technology shapes much of students' daily lives and gives them quick access to information. Students and teachers communicate increasingly through screens, changing how they interact. This shift has also brought problems such as cyberbullying, harassment, and emotional concerns that may go unnoticed.

Digital reporting tools can help schools organize student concerns and identify submissions that need attention. StudentShield processes vents and bullying reports submitted through its forms. It does not monitor private messages, social-media accounts, or other student communications.

The developers created StudentShield to support Saint Vincent Blessed School of Manila (SVBSM) in reporting and screening student concerns. Its counselor dashboard includes automated SMTP email and SMS notifications. These features support staff review, but their presence does not show that bullying or psychological harm has decreased.

**Purpose and Description**

StudentShield aims to help SVBSM identify and respond to emotional distress and reported bullying. Natural language processing screens student vents. Counselors investigate bullying reports and decide how to intervene.

Without a central monitoring system, schools may struggle to track student concerns and act before the situation becomes dangerous. StudentShield was developed around the following needs:

- Proactive Risk Detection: Use NLP and local language rules to screen student-submitted vents for signs requiring counselor attention.
- Priority Action: Classify vent submissions as Low, Medium, High, or Critical to help counselors prioritize review. Bullying reports follow a separate staff-verification workflow.
- Trend Analytics & Monitoring: Use wellbeing trends to identify patterns and help administrators make decisions based on student records.

**Project Description**

StudentShield combines a PHP CodeIgniter 4 web application with a Python FastAPI machine learning service. Students submit vents and incident reports through the web interface, and authorized staff review records through a unified dashboard.

Vent analysis assigns one of four risk categories: Low, Medium, High, or Critical. High vents prompt counselor notification attempts. Critical vents also prompt parent

notification attempts unless the student selects the parent-hiding safety option. Delivery depends on the configured services and connectivity. Bullying reports notify counselors at intake; staff must verify them before parent incident alerts are sent.

## Objectives of the Study

### General Objective

To design, develop, and evaluate StudentShield as a web-based emotional-distress screening and bullying-reporting system for SVBSM Junior High School students, helping guidance counselors review concerns, monitor risk levels, and coordinate follow-up.

### Specific Objectives

- To provide a private Safe Space and bullying-reporting website for SVBSM Junior High School students, with optional photo or video evidence.
- To classify student vents as Low, Medium, High, or Critical using mDeBERTa, GoEmotions, and language rules.
- To monitor changes in student risk levels using records from the previous 14 days.
- To send SMS and email alerts to counselors and eligible parents or guardians based on notification rules.

- To evaluate screening accuracy and privacy controls through a planned SVBSM pilot.
- To support one-to-one counseling through a new Google Meet link for each student session.

**Scope and Delimitations**

Scope

- Students sign in to submit vents and bullying reports, with optional photo or video evidence. Authorized staff can identify the students who submitted them.
- The system gives screening scores and risk levels. These are not clinically validated predictions of future harm.
- Risk monitoring uses up to 100 non-archived database records from the past 14 days.
- The staff dashboard supports SMS/email alerts: High vents notify counselors; Critical vents also notify eligible parents; verified incidents notify eligible parents.
- Vent and incident text normally use AES-256-CTR encryption. If vent encryption fails, the fallback stores unencrypted Base64.
- Registration requires acceptance of the terms and privacy information, followed by staff approval. Verified guardian consent is not recorded.

- Counselors publish Google Meet links. Calls run outside StudentShield, and the meeting host manages admission.

**Delimitations**

- The system does not diagnose mental illness, prescribe treatment, or replace professional counseling.
- It has no live chat or messaging module and does not host video calls.
- It is limited to SVBSM; multi-school and large-scale use have not been evaluated.
- An internet connection is required.

## Significance of the Study

The study aims to improve how SVBSM documents and prioritizes student concerns. The benefits below describe StudentShield's intended contribution, not measured intervention outcomes.

### The Client / Organization (Saint Vincent Blessed School of Manila)

The proposed study would give SVBSM a current tool for managing reports in one place. A centralized process would reduce unnecessary paperwork and verbal notifications, helping administrators make better use of guidance staff.

**End Users (Students and Guidance Counselors)**

Students can submit vents and bullying reports through a confidential channel. Submissions are linked to their accounts and remain accessible to authorized staff.

Guidance counselors can review the concerns together on the dashboard, where High and Critical vents are flagged for attention.

**The Parents**

The system can attempt SMS and email notifications to parents or guardians for Critical vents and staff-verified bullying incidents involving their children. The parent-hiding option suppresses the Critical-vent parent alert while keeping counselor escalation active. Parents receive notifications and have no login portal.

**The School / Institution**

StudentShield keeps reported concerns, staff actions, and case information in one place. Access restrictions and encryption support privacy requirements, but implementation alone does not establish legal compliance.

**The Proponents**

The team learned to connect a PHP and CodeIgniter web application with a Python and FastAPI microservice. This work involved combining web technologies, integrating APIs, using AES-256 encryption, and routing notifications in real time.

**Future Researchers**

The project shows how transformers can be used in a practical system. Future researchers could build on this work, including adapting the NLP engine to other dialects.

**Definition of Terms**

- Cyberbullying - Intimidation, harassment, or threats through electronic communication (Oxford Dictionary, 2020). In StudentShield, it refers to digital activity or aggression formally reported by SVBSM students through the "Bullying Report" module.
- Data Privacy Act (DPA of 2012 / RA 10173) - The Philippine law protecting the privacy of personal and sensitive information (National Privacy Commission). It provides the legal basis for how the system collects, stores, encrypts, and uses data.
- Encryption – Converting readable information into ciphertext to protect confidentiality. StudentShield normally encrypts vent and incident text with AES-256-CTR before storing it in MySQL. The vent controller's Base64 fallback remains a limitation.
- FAISS (Facebook AI Similarity Search) – A library for similarity searches over dense vectors. The earlier StudentShield design included FAISS, but the current database-backed risk-history function does not use it.

- Machine Learning – A field concerned with systems that learn patterns from data. StudentShield uses trained language models to screen submitted vent text. Microservice – A separately running service that provides a defined application function. The FastAPI text-analysis service is hosted separately from the CodeIgniter web application.
- Natural Language Processing (NLP) - A branch of artificial intelligence that analyzes, generates, and interprets human language (IBM, 2021). StudentShield uses NLP to analyze student submissions that contain slang and informal expressions.
- StudentShield – The capstone platform developed for SVBSM. It combines student vents, bullying reports, automated screening, staff case management, and eligible SMS/email alerts. The paper does not establish a sale or commercial transaction.
- Terms of Service (ToS) - The legal document describing user responsibilities, system limits, information handling, and interaction rules for students and administrators using the StudentShield website.
- Transformer - A neural network architecture that uses attention to connect words and sentences (Vaswani et al., 2017). In StudentShield, transformers will help interpret the meaning and context of student vents.

- Vent - To let off steam or express one's feelings (Merriam-Webster, 2023). In StudentShield, a vent is a student's text entry in the "Safe Space" module for expressing emotions.

# CHAPTER 2

# REVIEW OF RELATED LITERATURE AND STUDIES

## Related Literature

### A Review On Deep-Learning-Based Cyberbullying Detection. Future Internet

Hasan et al. (2023) review the use of convolutional neural networks, recurrent neural networks, and transformers to detect cyberbullying on social media. Their study reports that BERT and its variants consistently outperform other methods in identifying aggressive language without explicit slurs.

StudentShield also needs to recognize harmful language without relying only on explicit terms. This requires examining the full context of students' text.

Hasan et al. (2023) informed the team's choice of a transformer model, which is more complex than keyword and rule-based models.

### BERT-Based Ensemble Learning for Multi-Aspect Hate Speech Detection

Mazari et al. (2024) combined pre-trained BERT with Bidirectional LSTM and Bidirectional GRU networks for multiclass detection, reporting an ROC-AUC of 98.63%. Their model assigned categories such as "threat," "insult," and "identity hate," rather than only "toxic" or "not toxic."

StudentShield similarly needs to consider several risk factors in student vents, including threats and emotional expression. This study informed the decision to examine threat and emotion separately within the internal screening process.

**Natural Language Processing Applied to Mental Illness Detection**

Zhang et al. (2022) reviewed 399 studies on NLP for mental-illness detection using sources such as social media, interviews, and clinical notes. They identify data quality, language coverage, interpretability, and privacy as challenges. Their review describes research methods; it does not validate a clinical diagnosis from a student's vent.

The review is relevant because StudentShield receives student-written text that can contain Filipino, English, Taglish, and slang. Performance on another dataset cannot establish reliability for those submissions.

The project will therefore evaluate its current screening pipeline with an appropriate labeled test set and report class-level results. The public GoEmotions model is a separate emotion-model reference, and its labels must not be treated as a validated crisis assessment for SVBSM.

**DeBERTaV3: Improving DeBERTa with ELECTRA-Style Pre-Training**

He et al. (2021) proposed DeBERTaV3, an improved transformer that uses replaced token detection during training instead of masked language modeling. They report better scores and faster performance than BERT and RoBERTa, with fewer computing resources for text classification. Their benchmarks reached the top of the SuperGLUE leaderboard without expensive GPU clusters.

This work relates to StudentShield's need to run its analysis engine on conventional servers rather than data-center hardware. The school needs accurate, reliable screening at a low cost, with timely alerts.

The team drew on He and colleagues' findings when choosing DeBERTaV3 as the base transformer for StudentShield's threat-detection engine. The choice also considered the limits of typical school internet connections.

**Detection of Hate Speech using BERT and Hate Speech Word Embedding with Deep Model**

Saleh et al. (2023) trained BERT with custom word embeddings from annotated hate-speech datasets to recognize toxicity that was uncommon in generic training data. F1-score gains were larger for tweets containing covert hatred, where sarcasm or euphemisms obscured the meaning.

SVBSM Junior High School students use English, Filipino, and slang that the model may not have encountered during training. Fine-tuning is therefore needed to recognize coded language. Saleh et al. (2023) informed this approach to the classification engine.

**Bullying and cyberbullying: their nature and impact on psychological wellbeing**

Fermani et al. (2023) discuss the frequency, context, and effects of bullying and cyberbullying from several perspectives. Cyberbullying differs from traditional bullying because messages can remain visible indefinitely, reach a large audience, and allow perpetrators to remain anonymous. Stuart's studies report worse effects for students who experience both forms than for those who experience either one alone.

The team relates these findings to StudentShield's two reporting channels: Safe Space vents and the formal Bullying Report module. Together, these records help

guidance counselors determine whether a student is experiencing more than one form of bullying.

Drawing on Smith's research, the team chose to document both informal accounts of students' emotions and formal incident reports.

**BERT-based Ensemble Approaches for Hate Speech Detection**

Mnassri et al. (2022) report that stacking BERT-based models and voting on their results reduces false positives in hate-speech detection. Their study, presented at the IEEE Global Communications Conference, found that no single BERT configuration performed best in every case. Combining configurations can therefore improve prediction safety.

False alerts can damage a student's trust in a school reporting system. StudentShield needs to balance false positives against the risk of missing actual danger.

Mnassri et al. (2022) informed the team's plan to use mDeBERTa-v3 for threat detection and GoEmotions for emotion analysis, combining their results as a second opinion.

**Policy Guidance on AI for Children**

UNICEF (2021) sets out nine criteria for fair and transparent AI systems that protect information under relevant policies. It also calls for human oversight when the technology affects minors.

These recommendations support using technology alongside teachers' work. They caution against automating counseling, which could harm the students involved.

The team therefore proposes following UNICEF's human-in-the-loop approach to screen student vents and route eligible alerts. Counselors remain accountable for reviewing concerns and deciding the next action.

**Understanding the Influence of Cyberbullying Among High School Students in Philippine Public Schools in the Post-Pandemic Era**

Buguina and Buguina (2023) studied cyberbullying among high school students in the post-pandemic period. They found that some victims hesitate to report incidents and lack secure reporting channels, pointing to the need for structured reporting in Philippine schools.

Their study describes how cyberbullying affects students' wellbeing and academic performance. StudentShield addresses the reporting gap through an encrypted, confidential module. Students can speak up without fear of exposure, while guidance staff discuss their concerns before bullying escalates.

**Understanding Bullying in Philippine Education: Impacts and Opportunities for Change**

Communications and Communications (2025) discuss bullying in Philippine education and opportunities for change. They report a prevalence well above the OECD average, with nearly one in three Filipino students experiencing one or more forms of bullying. The lack of digital monitoring tools is cited as one reason schools struggle to intervene early. The team relates StudentShield to the institutional gap identified by

EDCOM The system gives SVBSM a way to monitor signs of harassment in student text and issue alerts before further harm occurs.

The EDCOM 2 report informed the team's plan for an early-warning platform that uses machine learning to identify harmful patterns in student writing.

**Revised Implementing Rules and Regulations of Republic Act No. 10627, Otherwise Known as the Anti-Bullying Act of 2013**

The Department of Education (2025) states in the updated implementing rules of the Anti-Bullying Act of 2013 that cyberbullying outside school and online harassment are covered. The guidelines ask schools to use digital tools to identify and document incidents and reduce manual work.

The team uses these rules as the legal framework for a digital monitoring platform. The policy also helps explain to parents and students why the school collects and processes vents through StudentShield.

The team designed StudentShield as a tool to support the school's obligations under RA 10627 by automatically logging, timestamping, and classifying flagged incidents.

**Emergency Alert and Warning Systems and Their Impact on Sustainable Disaster Preparedness and Awareness in the Philippines: A SEM–ANN Analysis**

Saflor and Kudhal (2026) studied perceived effectiveness of emergency alert and warning systems among 200 Philippine respondents. Their analysis links reliability,

message clarity, and trust with preparedness behavior; it does not establish that SMS is faster than email or guarantees delivery.

This relates to StudentShield's need to send important parent alerts promptly, including in areas with weak Wi-Fi. Relying only on online messaging may be insufficient for critical alerts.

The study informs StudentShield's alert design. Its SMS gateway attempts eligible parent alerts for Critical vent-screening results; successful delivery still requires testing.

**Anonymous Reporting Systems for School-Based Violence Prevention: A Systematic Review**

Messman et al. (2024) systematically reviewed anonymous reporting systems in United States schools. Only four relevant studies were identified, and one was published in a peer-reviewed journal. The authors called for more empirical evidence on effectiveness and noted differences in implementation and training.

The review is relevant to the design of a reporting channel, but it does not prove that such a channel will reduce bullying at SVBSM. A submitted report still needs staff assessment and appropriate follow-up.

StudentShield uses authenticated student accounts and offers privacy controls in its reporting workflow. This differs from an entirely anonymous reporting system. The proposed evaluation will assess whether students can use the reporting process and

whether staff can review its records; effectiveness should not be assumed from the presence of the feature.

**Mental Health Program Implementation of Local Universities and Colleges (LUCs) in Zambales: Basis for Action Plan Development**

Ebro and Mantillas (2024) examine the occurrence and forms of peer victimization in Philippine public secondary schools. They identify physical, verbal, social, and online abuse to provide a local framework for school administrators and educators responding to student hostility.

This work relates to how schools implement national anti-bullying policies. Ebro and Mantillas (2024) report that RA 10627 provides for anonymous bullying reports among junior students in public schools. Disciplinary action cannot be based on an anonymous report alone.

Anonymous reporting can increase reports from victims and witnesses, but each case still needs verification. Ebro and Mantillas (2024) support the need for data-management tools that help schools investigate these reports.

**Related Studies**

**Classifying the Severity of Cyberbully from Social Media Comments**

Nokham et al. (2025) analyzed 13,204 comments from Facebook, X, and TikTok, classifying them as Non-cyberbullying, Low, Middle, or High Severity. They compared BERT, Naïve Bayes, SVM, and stacking methods. The published abstract reports 94%

accuracy and an F1-score of 0.95 for SVM in binary classification. These are the study's results, not StudentShield's.

The study supports classifying levels of concern rather than using only a binary label. StudentShield applies this principle through four vent-screening bands and separate notification rules. The study does not validate StudentShield's thresholds or their clinical meaning.

**Protect: A Hybrid Deep Learning Model For Proactive Detection Of Cyberbullying On Social Media**

Harshitha et al. (2024) describe ProTect, a hybrid system that combines random-forest feature selection with a CNN to detect cyberbullying on Twitter and Instagram. The approach balances accuracy and speed because slow detection may allow harm to occur before an alert.

Speed also matters for StudentShield's vent screening, which can classify text in less than a second. In practice, a slow process may issue an alert too late.

This study informed StudentShield's fast text-classification approach, using a FastAPI microservice.

**A Multitask Framework for Sentiment, Emotion and Sarcasm aware Cyberbullying Detection from Multi-modal Code-Mixed Memes**

Maity et al. (2022) describe a multitask approach that considers sentiment, emotion, and sarcasm in cyberbullying detection. A trained algorithm can use emotion scores to assess the extent of cyberbullying in conversations.

StudentShield needs to distinguish urgent concerns from less urgent ones, rather than only deciding whether cyberbullying is present. Maity et al. (2022) informed the team's consideration of emotional-risk scores from 0 to 100 to help counselors prioritize students.

**Deep Learning in Sentiment Analysis: Recent Architectures**

Abdullah and Ahmet (2023) review deep-learning architectures for sentiment analysis. The paper provides background on methods for interpreting text, rather than evidence that StudentShield can predict a student’s future behaviour. It is not a study of SVBSM learners.

The review supports considering language-model methods for the screening component. StudentShield’s current trajectory feature instead uses stored risk labels and timestamps from the previous fourteen days. Its screening accuracy and the usefulness of its history view require separate evaluation; the review does not establish either outcome.

**Robust Hate Speech Detection in Social Media: A Cross-Dataset Empirical Evaluation**

Antypas and Camacho-Collados (2023) report that transformer models outperform conventional machine-learning methods in toxic-language detection, with further gains after fine-tuning on relevant datasets. StudentShield processes informal student text rather than news articles or Wikipedia entries. Its models therefore need training on text similar to social-media communication.

Antypas and Camacho-Collados (2023) informed the choice of models exposed to social-media-like text, including the informal language, abbreviations, and emojis used by SVBSM students.

**Tracking Emotional Shifts During Story Reception: The Relationship Between Narrative Structure and Affective Responses**

Schmidt et al. (2023) study emotions during short-story reception by comparing textual valence curves with retrospective ratings and physiological measures. Their work concerns narrative reception and does not validate StudentShield's crisis screening.

The study relates to the idea that emotional experiences change over time. StudentShield's risk-history assessment is a separate technical approach that needs its own validation.

The initial design proposed FAISS for comparing student submissions. The current approach assesses the sequence of risk labels from the previous 14 days.

**Different Interactions With Appearance-Focused Social Media Content And Adolescents' Body Dissatisfaction: A Within-Person Perspective**

Schreurs & Vandenbosch (2022) examine appearance-focused social-media interactions and adolescent body dissatisfaction using a three-wave panel design.

Their study does not examine whether a conversational reporting system increases student disclosure. The study provides background on adolescents' online experiences. It does not imply that every social-media interaction has the same psychological effect. Making Safe Space welcoming remains a usability goal that needs user evaluation.

**GoEmotions: A Dataset of Fine-Grained Emotions**

Demszky et al. (2020) introduced GoEmotions, a dataset of approximately 58,000 English Reddit comments annotated with 27 emotion categories and a neutral category. Each comment may receive multiple labels, allowing the dataset to represent overlapping emotional expressions. The researchers fine-tuned a BERT model and reported a macro-average F1-score of 0.46 across the emotion categories. Transfer-learning experiments showed that the dataset could support emotion classification in other domains and under different emotion taxonomies, although the results indicated that fine-grained emotion recognition remained challenging.

This study is relevant to StudentShield because it provides a foundation for analyzing emotions in student-submitted vents. StudentShield uses SamLowe/roberta-base-go_emotions, a RoBERTa classifier trained on the GoEmotions dataset, alongside its mDeBERTa distress model and multilingual screening rules.

Emotion scores such as sadness, fear, anger, and disappointment support the internal screening rules. The current API returns risk and trajectory results; the moods shown in stored vent entries are selected by students. The original study evaluated a BERT baseline, whereas StudentShield uses a separate RoBERTa implementation; therefore, the reported F1-score of 0.46 is not a performance result for StudentShield (Demszky et al., 2020; SamLowe, n.d.).

The study supports the use of fine-grained emotion analysis as one component of StudentShield's screening pipeline. However, GoEmotions was developed from English Reddit comments and did not establish performance on Filipino or Taglish student vents, nor did it validate clinical risk classifications. The selected classifier therefore requires evaluation using representative SVBSM submissions, while guidance counselors remain responsible for interpreting concerns and deciding on appropriate follow-up.

**Ensuring safety in digital spaces: Detecting code-mixed hate speech in social media posts**

Roy and Kumar (2025) report poor performance when hate-speech models classify code-switched messages that combine two or more languages in a sentence. Many toxic tweets were classified as neutral. Multilingual transformers performed better but still needed fine-tuning on examples from the target language pair.

SVBSM students often combine English and Filipino in Taglish sentences. StudentShield needs to recognize threats even when students switch languages within the

same message. Roy and Kumar's study informed the choice of mDeBERTa-v3, a multilingual transformer, to address code-switching in threat detection.

**Sentiment Analysis of Students' Experiences during Online Learning in a State University in the Philippines**

Santiago et al. (2023) describe an online sentiment-analysis system used to support counseling offices in Cavite. It collected short student messages and used NLP to classify their general mood. This saved counselors time and helped them identify declining mood over several weeks.

StudentShield adds transformer-based emotion and distress analysis with multilingual rules to interpret student concerns. It has not yet been shown to outperform Santiago et al.'s system.

The earlier study informed the addition of parent SMS alerts for Critical vent-screening results when notification rules permit parent contact.

**Empathy, Cyberbullying, And Cybervictimization Among Filipino Adolescents**

Shannen et al. (2021) study empathy, cyberbullying, and cybervictimization among 168 students in Grades 7 to 10 at a public high school in Cavite. This provides a Philippine adolescent setting for comparison. It does not establish SVBSM's prevalence, reporting patterns, or test anxiety.

The study by Shannen et al. (2021) relates to StudentShield's bullying-reporting requirement. A private reporting channel can help victims feel more comfortable sharing

concerns, since many are uncomfortable visiting a guidance office. The team therefore proposes a private channel where victims can report concerns without embarrassment.

**Views of Parents on Digital Parenting Competencies**

Hark Soylemez (2023) interviewed 13 parents about digital parenting. Their challenges included protection from online dangers and setting time limits. The study is not a survey of Philippine parents.

These findings relate to StudentShield's need to inform parents with different levels of technical skill. Notifications need to reach parents who do not have smartphones or internet access. Drawing on Hark Soylemez (2023), the team recommends automated SMS messages written in simple language.

**Awareness And Implementation Of The Anti-Bullying Law Among Secondary Schools In Northeastern Cagayan**

Pinera et al. (2022) report that schools in Central Luzon face heavy paperwork under RA 10627 because documentation is handled manually. Records accumulate in cabinets and are not always completed consistently. They recommend automating documentation where possible. StudentShield relates to this need by organizing reports and staff actions electronically. The study does not establish that SVBSM has the same compliance problems.

The team used the study to support structured incident records and audit logs. StudentShield does not automate the entire documentation process.

**Digital Literacy And Cybersecurity In Higher Education: The Unseen Power Of Academic Librarians**

Baharuddin et al. (2025) report that students readily recognize physical bullying but may overlook cyberbullying. Exclusion from discussions, passive-aggressive remarks, and anonymous insults were often treated as jokes.

These findings relate to StudentShield's goal of helping SVBSM students recognize cyberbullying. When students cannot identify an incident themselves, the software needs to help detect it. Drawing on Baharuddin et al., the team distinguishes four vent-screening categories from incident reports, which require separate counselor verification.

**Online Counselling Among Filipinos: Do Internet-Related Variables Matter?**

Tuliao and Velasquez (2017) examine online counseling among Filipinos and internet-related factors. Their work relates to online help-seeking but does not establish acceptance of StudentShield's interface.

The review relates to Safe Space because students need to trust the feature and be confident that their statements will not appear in their academic records. Safe Space is separate from systems for grades and attendance. StudentShield stores profile details such as grade and section but does not maintain academic records.

**Improving Large-scale Language Models and Resources for Filipino**

Cruz and Cheng (2022) describe adapting pre-trained models from a resource-rich language such as English to Filipino, which has fewer training resources. Multilingual BERT models fine-tuned for Filipino performed significantly better than models trained from scratch on a small dataset.

This approach relates to StudentShield's Taglish text, which mixes English and Filipino. Training a model only on Filipino data may be ineffective because the available training data is limited. Drawing on Cruz and Cheng, the team proposes using a pre-trained multilingual transformer, mDeBERTa-v3, to recognize language patterns in sentences.

**Synthesis of the Reviewed Literature and Studies**

The reviewed work addresses two related needs: interpreting student language and providing a way to report concerns. Classification research informs the screening component, while reporting-system research points to the need for staff follow-up. These studies do not establish the accuracy, usability, or effectiveness of StudentShield. In particular, Messman et al. (2024) found limited empirical evidence for anonymous reporting systems, so a reporting feature alone cannot be treated as proof of improved school safety.

Philippine studies provide context for bullying, help-seeking, and language use, but their participants and settings differ from SVBSM. Shannen et al. (2021) and Tuliao

and Velasquez (2017) inform the discussion of adolescent experiences and online counselling. Cruz and Cheng (2022) show the value of developing Filipino language resources. These sources support testing with the intended language and users; they do not supply SVBSM prevalence figures or validate the current model. For this project, the remaining question is how the components work together in the school's actual workflow. StudentShield combines private student submissions, staff review of incident reports, text screening with a local fallback, stored risk history, conditional notifications, and assigned Google Meet session links. The proposed evaluation will examine this combined workflow and its limits. Model performance, notification delivery, and user acceptance remain separate outcomes that require test evidence and participant feedback.

# CHAPTER 3

# TECHNICAL BACKGROUND

## Overview of Current Technologies

StudentShield separates the web application from machine-learning analysis. This design is intended to keep student submissions from failing and give the ML engine enough computing resources to analyze several texts at once. The separation matters during busy periods, such as exams, when student use increases sharply.

- Front-End Student and Administration Portal: The interface uses HTML5, JavaScript, and Tailwind CSS. These technologies keep pages simple and help them load quickly.
- Back-End Application Logic (PHP): CodeIgniter 4 handles routing, sessions, database access, and workflows. The application enables the global CSRF filter and role-based access restrictions. These controls do not guarantee that the application has no vulnerabilities.
- Database (SQL): MySQL/MariaDB stores user profiles, submissions, incidents, and case records. CodeIgniter's AES-256-CTR encrypter normally protects vent text and incident details, except for the vent-controller fallback described in the delimitations.
- Machine Learning Analysis Engine (Python): FastAPI and PyTorch run text analysis. A separate deterministic function calculates 14-day risk history from records supplied by MySQL.

- Advanced Language Models: The emotion pipeline uses SamLowe/roberta-base-go_emotions. The distress pipeline first tries the locally fine-tuned model, then ashlogg/student-shield-mdeberta on Hugging Face Hub. If neither is available, it uses GoEmotions. The repository includes training scripts and datasets. Collecting 300 counselor-labeled SVBSM entries and expanding the set to 1,000 remains a proposal; the paper does not document its completion.
- Trajectory Tracking: VentController collects the current submission and risk labels and timestamps from the previous 14 days. risk_history.py orders the records, adds the current preliminary classification, and calculates trend and recurrence scores. History comes from the database; the ML service has no per-student FAISS file.
- Deployment and Notification: InfinityFree hosts the CodeIgniter web application, and a Docker Space on Hugging Face hosts the FastAPI service. Cloudflare serves the web domain. Environment variables configure the SMS gateway and SMTP service. The production PHP host has no SSH access.

**Risk Classification Engine**

**StudentShield Risk Classification Engine**

The engine gives equal weight to distress and negative-emotion confidence, then applies additive language rules and ordered overrides. It checks historical escalation using 14 days of database-backed risk labels. The returned percentage is limited to its

final Low, Medium, High, or Critical band. These scores are not validated probabilities of future clinical outcomes.

| Stage | Current implementation | Interpretation |
|---|---|---|
| Base score | 50% distress confidence + 50% negative-emotion confidence | The current base formula does not use a 30% semantic-similarity component. |
| Additive heuristics | Hopelessness +0.15; masked distress +0.14; emotional distress, isolation and self-worth +0.12 each; distortion and relationship distress +0.10; school avoidance +0.18. | These software weights have not been independently validated clinically. |
| Compound signals | Adds 0.15 × (signal count − 1) when at least two selected indicators match. | This is an additive increase, not an exponential multiplier. |
| Ordered overrides | Explicit self-harm, passive-disappearance ideation, finality, violence, sexual harassment, emotional and contextual rules. | Passive-disappearance matches set a 0.79 High floor before later adjustments. Other matching rules may change the output. |
| Risk history | Risk labels and timestamps from 14 days, ordered with the current label added for trend and severe-recurrence checks. | MySQL records replace the earlier FAISS recurrence store. |
| Output bands | Low 1–30%; Medium 31–60%; High 61–85%; Critical 86–100%. | Scores are limited to the final risk band and are not validated probabilities of future harm. |

**Table 1.** ***Current risk-screening stages***

**Evaluation of the Existing System (Manual or Automated)**

During consultations with SVBSM, the proponents were informed that student concerns were handled through traditional, manual reporting and follow-up procedures. The school reported concerns involving bullying and suicidal thoughts among high school students, some of which were not addressed promptly. These reported difficulties identified a need to organize concerns and support timely counselor review and follow-up. The study has not established baseline processing times, reporting rates, or intervention outcomes for comparison.

**Technical Feasibility**

**Hardware Specifications**

The specifications below come from the original development proposal. They have not been benchmark-tested as minimum requirements for transformer inference or production hosting.

| Component | Proposed baseline | Preferred / testing provision |
|---|---|---|
| Processor | Intel Core i3 / AMD Ryzen 3 for web development | The proposal did not specify a separate recommended processor. |
| Memory | 4 GB RAM in the original plan | 8 GB RAM in the original plan |
| Storage | 128 GB SSD | 256 GB SSD |
| Graphics | Integrated graphics for browser and PHP work | These web-development specifications do not cover ML training hardware. |

| | | |
|---|---|---|
| Operating system | Windows for the Laragon workflow | Use a supported environment compatible with the current PHP and browser tools. |

**Table 2.** ***Proposed hardware specifications***

The original SMS gateway proposal specifies an Android 8.0+ phone with an active SIM and network connection. Current code calls a configured gateway endpoint, but the supplied materials do not establish the deployed device or its delivery capacity. Browser tests will use a desktop or laptop and a smartphone to check student forms and staff dashboards.

**Software Specifications**

The table distinguishes development tools from configured application and hosting components. Unconfirmed versions and delivery capacities are not measured production requirements.

| Component | Configured use |
|---|---|
| Web stack | PHP 8.1+ and CodeIgniter 4; HTML, CSS, Tailwind CSS, and JavaScript. |
| Development | Laragon, a code editor, and a current desktop/mobile browser. |
| Database | MySQL / MariaDB for accounts, submissions, case history, and sessions. |
| Charts | Chart.js and ApexCharts for dashboard and student-history views. |

| | |
|---|---|
| ML service | Python 3.10+, FastAPI, and PyTorch in a Docker-based Hugging Face Space. |
| Hosting | InfinityFree for PHP; Cloudflare HTTPS for studentshield.app; Hugging Face Spaces for ML. |
| Notifications | Environment-configured SMTP and SMS gateway; verify delivery separately. |
| Security | Password hashing, CSRF controls, encrypted vent text, and six-digit staff email OTP. |
| Sessions | Staff publish Google Meet links; assigned participants join through the application's authorized redirect. |

**Table 3.** ***Software and deployment specifications***

CodeIgniter's AES-256-CTR encrypter normally protects vents; its error fallback remains a limitation to test. Submission cooldowns are 30 seconds for vents and 120 seconds for reports. Optional trusted-device cookies can bypass staff OTP verification. The production PHP host has no runtime CLI or SSH access, so deployment and database changes must use compatible hosted workflows.

**Network/Architecture Specifications**

Test Network: Browsers need access to the web application, and the application host must reach the configured SMS endpoint. End-user devices do not all need to share the gateway's local Wi-Fi network.

The web interface and ML engine have separate hosting and maintenance. The web application handles accounts, stored submissions, incidents, and staff actions. The

Python service returns screening results. When no ML result is available, the PHP fallback provides reduced screening.

StudentShield has four authenticated roles: Student, Counselor, Super Admin, and Principal. Students submit entries and reports. Counselors review cases, approve registrations, and manage counseling sessions. Super Admin manages accounts and audit records, while Principal views aggregate school reports. Parents or guardians receive eligible notifications without a portal. CodeIgniter 4 and MySQL run on InfinityFree; FastAPI and PyTorch run on Hugging Face Spaces. Vent-analysis requests use X-API-KEY authentication. Assigned participants join counselor-published Google Meet links through the browser.

**Software Architecture**

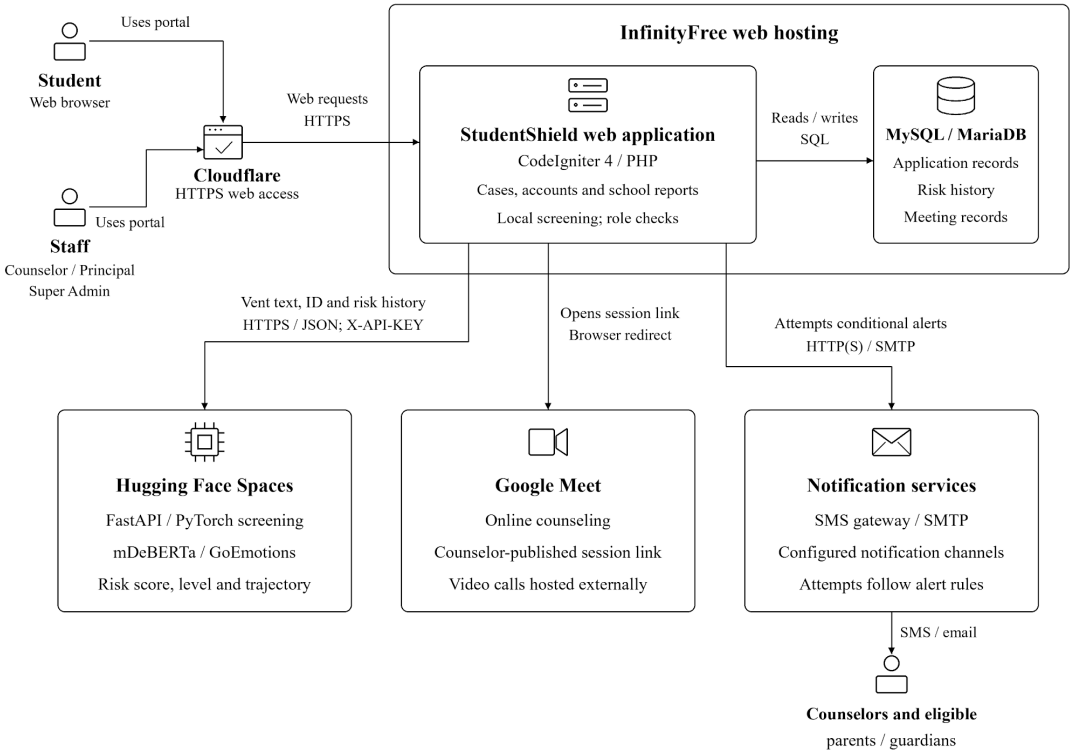


**Figure 1.** ***Software Architecture***

Figure 1 shows browser access for four roles through Cloudflare to CodeIgniter 4 and MySQL on InfinityFree. FastAPI on Hugging Face Spaces receives vent text, an internal student identifier, and recent risk history, then returns screening and trajectory results. PHP retains local safety screening. Configured SMS/email services attempt eligible notifications. Counselors publish Google Meet links; StudentShield checks participant access before redirecting to the external call.

# CHAPTER 4

# METHODOLOGY, TESTING, AND EVALUATION

## Software Development Life Cycle (SDLC Model)

StudentShield uses an Agile-Iterative approach. Figure 2 shows the cycle of requirements, design, development, testing, deployment, and review. Each cycle addresses selected features, with test findings and feedback guiding revisions. Planning and analysis define requirements; maintenance follows deployment and review.

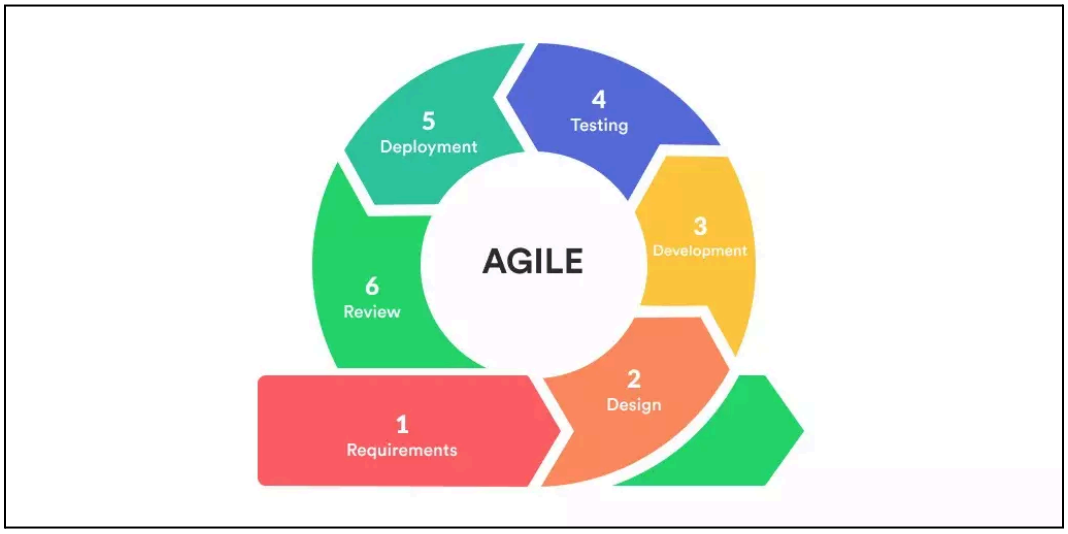



**Figure 2.** ***Agile-Iterative Development Model***

- **Planning**

  Planning identifies the needs of SVBSM students and guidance staff and sets the scope of each cycle. Work is prioritized around Safe Space, bullying reports, screening support, and counseling sessions.

- **System Analysis**

  System analysis defines student submissions, counselor case review and registration approval, Super Admin account administration, and Principal aggregate reporting. It also identifies access restrictions, privacy needs, alert conditions, and the data exchanged with the screening service.

- **System Design**

  System design turns the requirements into page layouts, data structures, and process flows. It defines how submissions reach the dashboard, how screening results appear, and how students request and join counseling sessions.

- **Development**

  Development has three areas, each revisited during testing and revision:

  Iteration 1: Core web features, including account access, vent submissions, bullying reports, and staff review.

  Iteration 2: Integration of the FastAPI screening service on Hugging Face Spaces and the local screening fallback.

  Iteration 3: Security, notifications, and counseling support. This covers vent encryption, staff email verification, role restrictions, IP audit records, configurable SMS/email alerts, and counselor-published Google Meet links. The application uses CodeIgniter 4 and MySQL, with Tailwind CSS and JavaScript for the interface.

- **Testing**

  Testing covers the following areas before changes are released:

  Account access, staff email verification, submission handling, record storage, and permissions for each user role.

  Risk and emotion screening using labeled English, Taglish, slang, and regional-language samples, including the local fallback when the screening service is unavailable.

  Alert handling under different risk levels and notification preferences, including missing gateway settings and delivery failures.

- **Deployment, Review, and Maintenance**

  InfinityFree hosts the web application, and Hugging Face Spaces hosts the screening service. Issues and feedback after each release guide the next cycle.

  Maintenance includes:

  Checking the web application, database connection, screening service, and notification settings after deployment.

  Reviewing student and staff access permissions as system requirements change.

  Correcting reported errors and improving page layouts and navigation.

  Revising screening models and language rules when evaluation results identify errors.

  Applying security updates and maintaining system backups.

**Implementation Plan**

The original Gantt chart remains a proposed schedule, not evidence that activities are complete. Its title states February to May, but its columns span Month 1 to Month 10.

| Activity / month | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Planning Phase | | | | | | | | | | |
| Defining the Problem | ■ | | | | | | | | | |
| Confirming Project Feasibility | ■ | | | | | | | | | |
| Producing the Project Schedule | ■ | | | | | | | | | |
| Chapter 1, 2, 3, 4 | | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | |
| Title Defense | | ■ | | | | | | | | |
| Prototype Defense | | | | ■ | | | | | | |
| Analysis Phase | | | | | | | | | | |
| Interviewing the Target Beneficiary | ■ | ■ | | | | | | | | |
| Consulting a Mental Health Professional | | | ■ | | | | | | | |
| Determining System Requirements | | | ■ | ■ | | | | | | |
| Determining User Requirements | | | ■ | ■ | | | | | | |
| Recommending Solutions | | | ■ | ■ | | | | | | |
| Designing DFD, CFD, ERD | | | ■ | ■ | | | | | | |
| Design Phase | | | | | | | | | | |
| Determining the Programming Tools | | | | ■ | ■ | ■ | ■ | ■ | ■ | |

**Table 4.** ***Original proposed implementation schedule (month indices) (continued)***

| Activity / month | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Designing the Database | | | | | | | | | | |
| Designing the System Interface | | | | | | | | | | |
| Designing ML System Control | | | | | | | | | | |
| Implementation Phase | | | | | | | | | | |
| Revising the Document | | | | | | | | | | |
| Coding | | | | | | | | | | |
| Testing and Verifying the System | | | | | | | | | | |
| Mock Defense / Final Checking | | | | | | | | | | |
| Final Proposal Defense | | | | | | | | | | |

**Table 5.** ***Original proposed implementation schedule (month indices)***

**Requirements Analysis**

The requirements were informed by consultations with SVBSM about its existing procedures for handling student concerns. The school described traditional, manual reporting and follow-up and reported bullying and suicidal concerns among high school students that were not always addressed promptly. These accounts describe needs reported by the school; they are not prevalence estimates or independently verified clinical findings.

These needs guided the requirements for a private student reporting channel, centralized concern records, screening to help counselors prioritize vents, and SMS/email notifications to support follow-up. Bullying reports require counselor verification, while screening results support review rather than diagnosis. Authorized school personnel remain responsible for assessing concerns and deciding on intervention. The functional and non-functional requirements below describe how StudentShield supports these tasks.

**Functional Requirements**

**Role-Based Access Control**

- Users have four roles: Student (role 0), Counselor (role 1), Super Admin (role 2), and Principal (role 3).
- Students can open their dashboards, submit vents and bullying reports, and request or check counseling sessions.
- Counselors review vents, incident reports, and student risk profiles; approve or reject registrations; verify or reject incidents; record case notes and status; and manage counseling sessions.
- Super Admin manages accounts, assigned roles, access restrictions, and audit logs. This role cannot open clinical case records, student profiles, or counseling sessions.
- Only Super Admin can administer accounts. Counselors, Principals, and Students cannot use those routes. Principals filter and export aggregate CSV summaries; individual case records remain restricted.

**Account Management**

- Counselor, Super Admin, and Principal accounts are added by Super Admin. Each staff role has an account-name menu for Profile, Settings, and Logout.
- Student self-registration creates a pending, inactive account for counselor approval. Super Admin can create a student account, but it remains pending until a counselor approves it. Account enabling does not approve a registration.
- Self-registration collects a twelve-digit Learner Reference Number (LRN), student profile details, and available parent or guardian contacts. The LRN must be unique. Users sign in with their account name and password rather than the LRN or generated student ID.
- Counselors, Super Admins, and Principals sign in through the staff portal with their name and password, followed by a six-digit email OTP. The OTP expires after five minutes and has an attempt limit. A valid remembered-device token can bypass a new OTP challenge.
- Students sign in with their name and password. They do not need an OTP.
- Passwords are stored as bcrypt hashes and checked with password_verify. The paper does not claim automatic upgrades of plaintext passwords.
- There is no password-reset feature. Super Admin manually sets a forgotten password through user management.

- Self-registration sets registration_status to pending and is_active to 0. Login also checks access_disabled. Account changes increment auth_version and clear remembered access so that outdated sessions can be rejected.
- Registration requires acceptance of the terms and privacy information shown on the page.
- The sign-up process requires acceptance of terms and privacy information but does not separately verify or store signed guardian consent. Pending approval is not verified guardian consent.

**Vent Log Submission and AI Risk Analysis**

- Students submit vents with an expandable, optional feelings chooser.
- When configured, the application requests analysis from the FastAPI service on Hugging Face Spaces. A usable response supplies a Low, Medium, High, or Critical risk level.
- The PHP client allows 15 seconds to connect and 60 seconds for the full ML request. If no usable risk response arrives, local multilingual keyword screening runs instead. Explicit suicide phrases can override the returned category. Context and sarcasm rules are screening aids, not guaranteed interpretations.
- Vent text normally uses AES-256-CTR encryption and hex encoding before storage. If encryption fails, the current fallback uses Base64 encoding, leaving a confidentiality limitation.

- Critical vents trigger counselor notifications and eligible parent SMS/email attempts unless parent hiding is selected. High vents trigger counselor-only notifications.
- Selecting "hide from parents" suppresses parent alerts and sends an emergency override notification to the counselor. The option is a precaution for domestic abuse.
- A visible privacy notice appears directly below the "Submit" button to explain how student data is handled.
- The notice describes confidential handling and encryption. The normal storage path is encrypted, but encryption cannot guarantee that disclosures or leaks will never occur.
- Students are told about conditional parent alerts for Critical concerns. Authorized staff can review submissions at every risk level; confidentiality does not mean anonymity from staff.
- The parent-hiding option suppresses the Critical-vent parent alert and notifies the counselor. The code does not require a confirmed domestic-abuse finding to honor the selection.

**Bully Report Submission (Two-Stage Alert Pipeline)**

- Students submit encrypted bullying reports and identify whether they are the victim or reporting for someone else.

- The system resolves typed names to database user IDs only when the lookup returns one match, to prevent wrong-parent notifications.
- Stage 1 (Report Intake): Submission prompts SMS and email notification attempts to configured counselor contacts. No parent bullying-incident alert is sent at this stage.
- Stage 2 (Staff Verification): Verifying a report prompts alert attempts to the resolved victim's and alleged aggressor's parent contacts. Staff can also reject a report or undo a decision. Rejected reports add progressive strikes: the second can prompt a parent alert, and three or more lock the reporter's account until staff act.

**Dashboard and Case Management**

- Counselors can review all active vents with decrypted text, risk scores, student names, sections, and adviser details.
- The dashboard's risk-distribution pie chart (Chart.js) and vent-trend area chart (ApexCharts) update in real time through AJAX polling.
- Counselors can resolve entries, archive them through soft deletion, restore them, or apply bulk actions.
- Selecting a student name opens that student's profile, risk timeline (ApexCharts), full vent history, and involvement in bullying reports.
- Super Admin Account Management supports creating, editing, and deleting users, assigning roles, and reviewing audit records.
- Super Admin cannot demote or delete their own account.

- Students can request one-to-one counseling. Counselors create a new Google Meet externally, confirm host and admission settings, and publish the link through the case summary. The assigned student's Join control redirects to Google Meet. Sessions are not limited to Critical-risk students.
- Counselors can open a Case Summary Report for an individual student.
- The report combines risk history, incidents, notification history, counselor notes, and case-status history.
- The view brings together records for counseling follow-up and incident investigation.
- The case summary includes print styles for browser printing or saving as PDF.

**Automated Notifications**

SMS alerts use a configured gateway. The web application sends the recipient's phone number and a prepared message to the endpoint set as sms.GatewayUrl. The application host must be able to reach that endpoint. If an Android phone serves as the gateway, it needs an active SIM, an available SMS allowance or balance, and mobile signal. A private local-network address alone is not reachable from the hosted application; the gateway needs an appropriately secured, reachable connection.

High vents (61–85%) prompt counselor-only SMS and email attempts. Critical vents (86–100%) prompt counselor alerts and eligible parent or guardian SMS/email

attempts, subject to available contacts and the parent-alert cooldown. Selecting hide from parents suppresses the Critical-vent parent alerts and prompts an additional counselor notice. Missing parent contacts are indicated in the counselor alert.

Bullying-report intake prompts counselor SMS/email attempts so the report can be investigated. Parent incident alerts are attempted only after counselor verification, using available contacts for the resolved victim and alleged aggressor records. Separate receipt and false-report notices follow their own workflow and must not be confused with verified-incident alerts.

SMS messages provide a notification rather than the full vent or incident narrative. Recipients do not need a StudentShield portal account to receive SMS. The gateway requires connectivity for the application's request, while the recipient needs a working mobile number and cellular service. Email uses separately configured SMTP; it is not a guaranteed fallback triggered by failed SMS delivery.

Eligible parent vent alerts check the recipient's cooldown; this check does not apply to every notification type. Vent-alert attempts are recorded in a JSON file before dispatch. A recorded attempt or gateway response does not establish handset delivery or that the recipient read the message. Error reporting varies across notification paths, and the implementation does not provide verified delivery receipts or a guaranteed automatic retry. Counselor follow-up remains necessary when an alert is unavailable or unacknowledged; delivery and failure handling will be evaluated during testing.

**Audit Logging**

- The audit_logs table records supported administrator and case actions, including the acting user, action, target, IP address, user agent, and timestamp. Recording these actions does not establish that audit-completeness testing is finished.

**Data Retention Policy**

- The original paper proposes keeping records until a student graduates or transfers, then for another 30 days. This is a proposed school-wide policy, not an implemented automatic-retention process.
- The proposed 30-day period allows pending administrative and counseling matters to be completed.
- Automatic deletion after this period has not been implemented or verified in the paper. The study does not establish the proposed period as a statutory requirement.
- The original proposal also preserves records during inactivity and restores them upon re-enrollment.
- Vent and incident archives use soft-delete fields. These fields alone do not implement enrollment-based retention.
- Enrollment-based archiving, timed deletion, and restoration on re-enrollment still need implementation and testing.
- Only authorized school personnel may access archived records.

**Non-Functional Requirements**

**Performance**

- The original document proposes a 2 to 3 second vent-response target, but provides no benchmark. Synchronous ML and notification requests may lengthen submission time.
- ML requests have a 60-second total timeout, including 15 seconds for connection. Local keyword screening provides a fallback when the request fails. Fallback response time has not been benchmarked.
- AJAX polling will refresh dashboard data without reloading the whole page.

**Availability**

- InfinityFree hosts the application, while Hugging Face Spaces hosts the ML model separately. The PHP fallback reduces reliance on the ML service. Neither service guarantees continuous availability or sub-second response times.

**Security**

- Vent and incident text normally use AES-256-CTR during storage. The vent Base64 fallback and plaintext-reading support prevent a claim that all stored text is encrypted.
- Passwords are hashed with bcrypt and checked with password_verify.
- Staff verification uses six-digit OTPs with a five-minute expiry and an attempt limit. Remembered devices follow the exception described above.

- Submission limits of 30s for vents and 120s for bullying reports help prevent spam and gateway misuse.
- Only Super Admin can access user management. Counselors are blocked.
- Automatic retention remains unimplemented. There is no code for automatic anonymization at the end of the school year or thirty days after transfer.

**Usability**

- The Tailwind CSS interface adapts to desktop and mobile devices.
- Students submit a vent in two steps: enter the text and press Submit.
- The staff dashboard shows active cases and retrieves updated records through asynchronous browser requests. Its usability with large record volumes has not been studied in the paper.

## Conceptual Framework

The Input-Process-Output (IPO) framework shows how user needs, system data, and development resources guide StudentShield's development. It also includes evaluation to identify improvements.

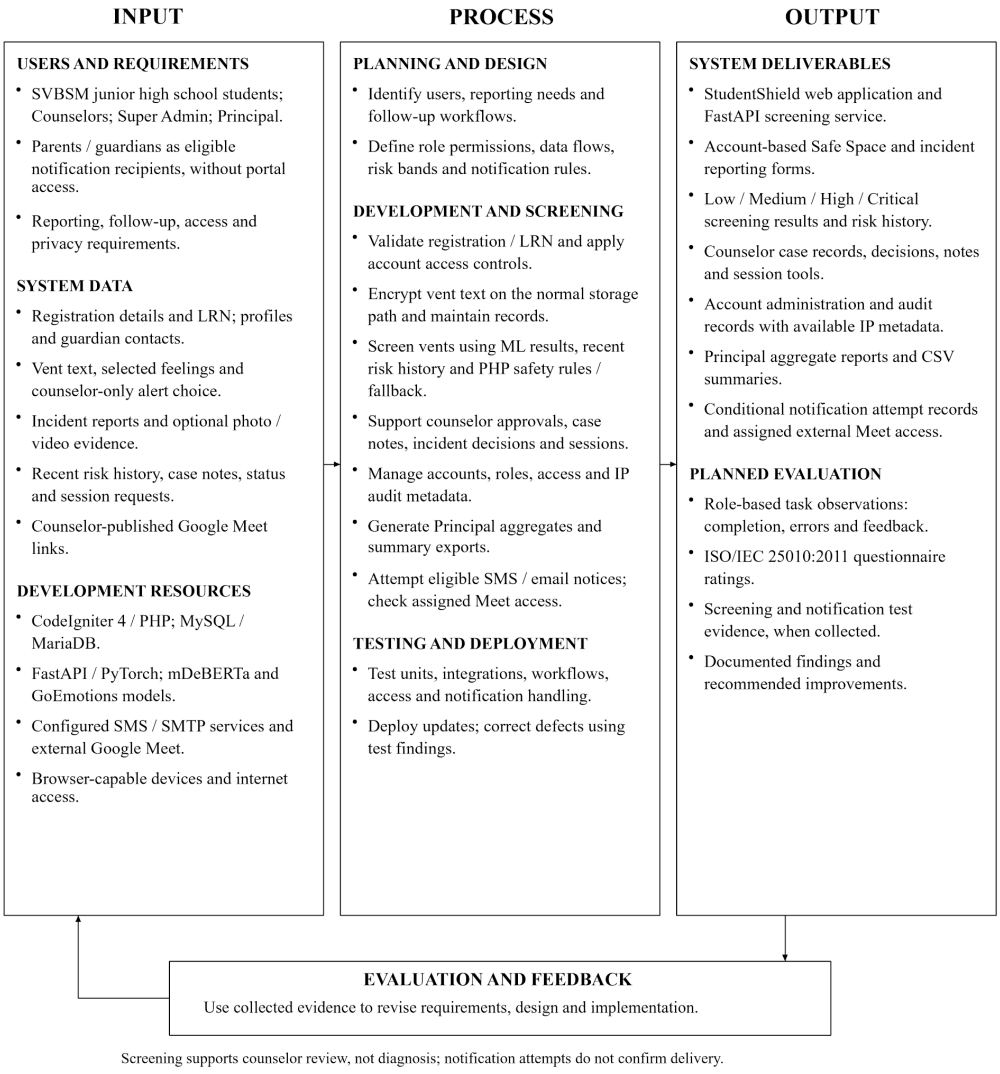



**Figure 3.** ***Conceptual framework***

Figure 3 presents StudentShield's inputs, development processes, and outputs. Evaluation and feedback connect the resulting features and collected evidence to revisions in requirements, design, and implementation.

**Input**

Inputs comprise SVBSM user and privacy requirements; registration details and LRN; profiles and guardian contacts; vents, feelings, and alert preferences; incident reports and optional evidence; risk history, case records, session requests, and counselor-published Meet links. Resources include CodeIgniter 4/PHP, MySQL/MariaDB, FastAPI/PyTorch, mDeBERTa and GoEmotions, configured SMS/SMTP services, external Google Meet, browser devices, and internet access.

**Process**

Planning and design establish reporting workflows, role permissions, data flows, risk bands, and notification rules. Development implements registration and LRN validation, access controls, normal-path vent encryption, ML screening with recent risk history, and PHP safety rules and fallback. Counselor case and session workflows, account administration, IP audit metadata, Principal aggregates, conditional notification attempts, and assigned Meet access are integrated. Unit, integration, workflow, access, and notification tests guide deployment and corrections.

**Output**

Outputs include the StudentShield web application and FastAPI screening service; Safe Space and incident-reporting forms; Low, Medium, High, or Critical screening results and risk history; counselor case and session tools; account administration and audit records with available IP metadata; Principal aggregate reports and CSV summaries; conditional notification-attempt records; and assigned external Meet access. Screening supports counselor review rather than diagnosis, and notification attempts do not confirm delivery.

**Evaluation**

The planned evaluation combines role-based task observations, including completion, errors, and feedback, with ISO/IEC 25010:2011 questionnaire ratings and screening and notification test evidence. Findings will be reported after the corresponding activities are completed and will guide revisions to requirements, design, and implementation.

### 4.3 System Design and Modeling

StudentShield's system design translates the requirements into user workflows, data flows, and database relationships. The context diagram defines the system boundary and external entities. The use case diagram identifies the actions available to each actor, while the data flow diagram describes how submissions, screening results, and records

move through the system. The entity relationship diagram and data dictionary describe the database tables, relationships, keys, and fields.

The interfaces connect users to these processes through role-based access. Students submit Safe Space entries and incident reports and request counseling sessions. Counselors review registrations and cases, record notes, update statuses, and publish assigned session links. Super Admin manages accounts and access and reviews audit records. The Principal views aggregate analytics and exports summaries. Parents or guardians receive eligible notifications without a portal account.

The CodeIgniter web application handles browser requests and database operations. The FastAPI service analyzes vent text and returns screening results for counselor review, with local PHP safety rules providing a fallback when the service is unavailable. The following diagrams explain how the interfaces, processing logic, and stored data work together.

### 4.3.1 Context Diagram

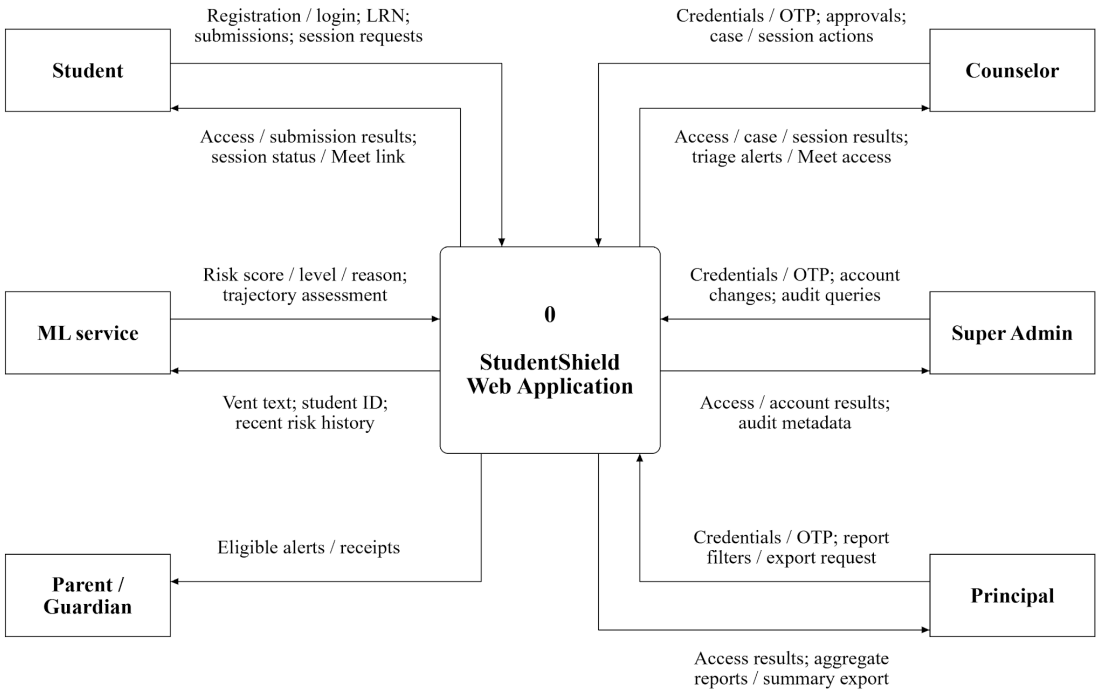



**Figure 4.** ***StudentShield context diagram (DFD Level 0)***

Figure 4 shows one system process and six external entities. Student supplies registration details, LRN, and submissions. Counselor supplies approval, case, and session actions. Super Admin supplies account changes and audit queries. Principal supplies report filters and receives aggregates. The ML service receives vent text, an internal student identifier, and risk history and returns screening and trajectory results. Parents or guardians receive eligible alerts and receipts without portal access. Meeting links are published by counselors and opened through an authorized browser redirect.

### 4.3.2 Use Case Diagram

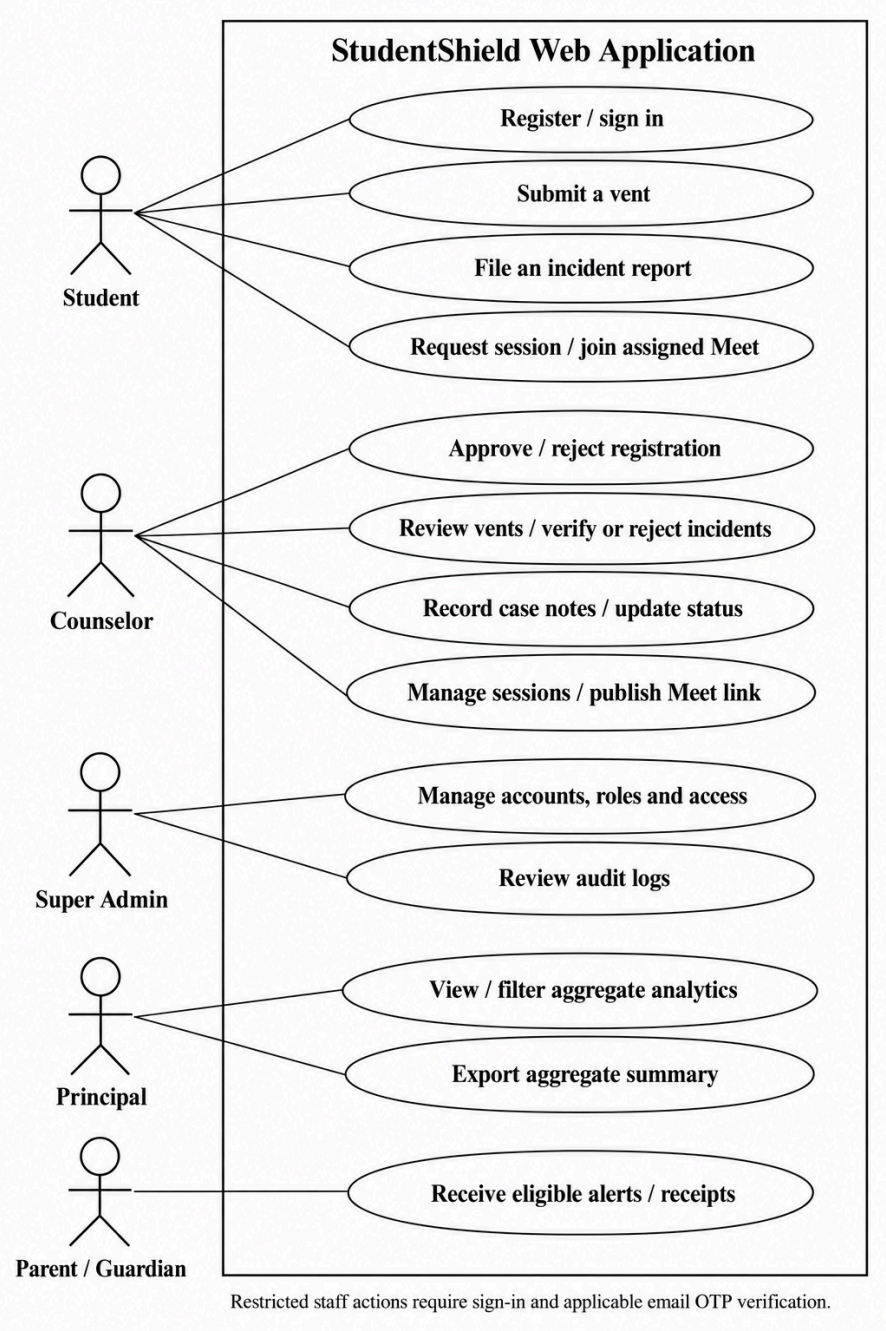



**Figure 5.** ***StudentShield use case diagram***

Figure 5 maps the goals available to the four authenticated roles and the notification-recipient role. Staff authentication is a precondition for restricted actions. Each association identifies an actor that participates in the corresponding use case.

**Actors**

- Student: Registers, submits vents and incident reports, and requests or joins an assigned counseling session.
- Counselor: Approves registrations, reviews cases, records notes and status, and manages counseling sessions.
- Super Admin: Manages accounts, roles, access, and audit metadata. Clinical records and sessions are excluded.
- Principal: Views, filters, and exports aggregate school reports. Individual clinical records are excluded.
- Parent/Guardian: Receives eligible alerts and receipts without signing in. A notification attempt does not confirm delivery.

**Access and notification rules**

Restricted staff actions require password authentication and applicable email OTP verification. Students do not use email OTP. High vents prompt counselor-only alert attempts; Critical vents can also prompt eligible parent alerts unless the student suppresses them. Separate parent incident alerts follow counselor verification. Report receipts use the stored contact address. Counselors verify reported incidents before treating them as misconduct.

### 4.3.3 Data Flow Diagram (DFD)

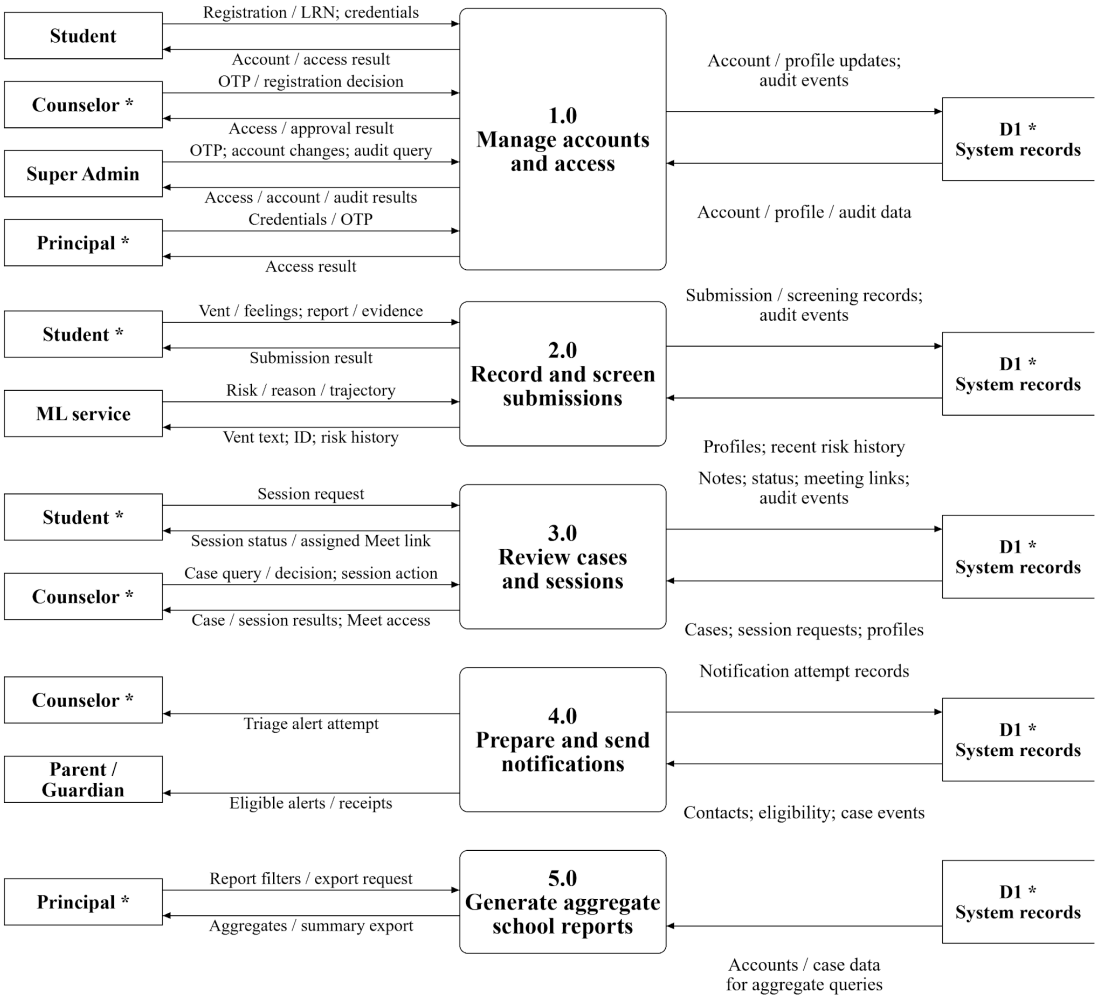
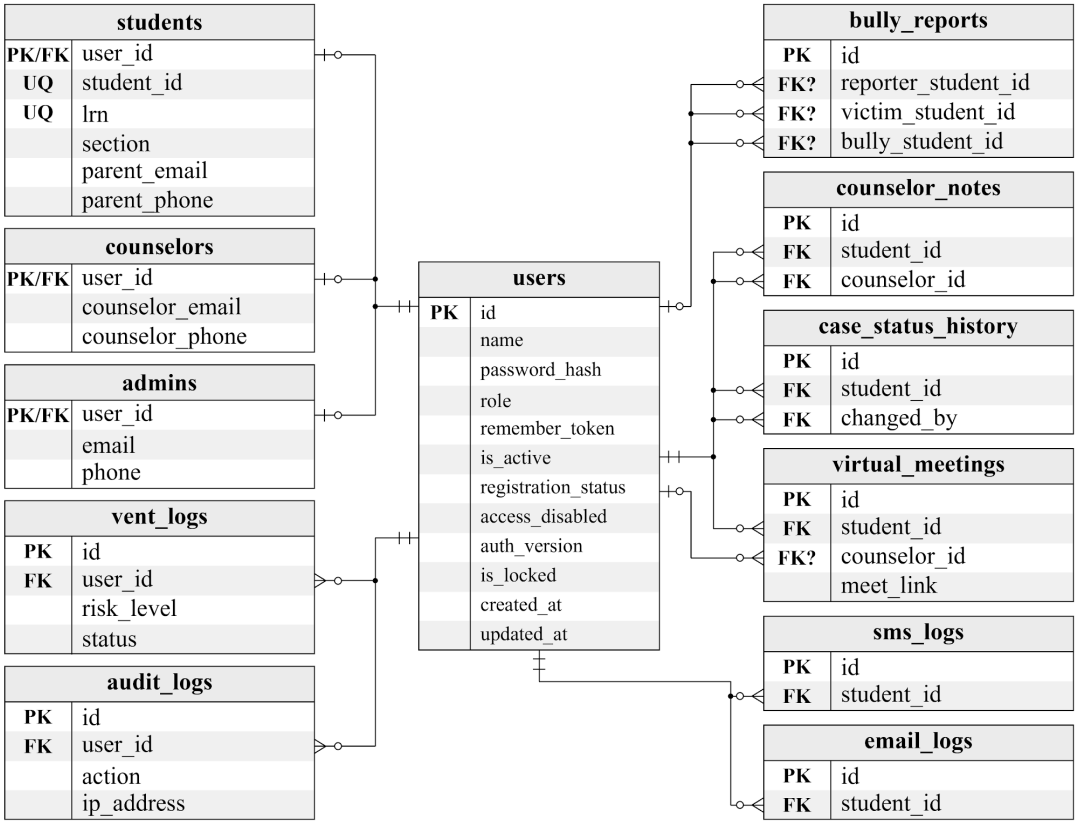



**Figure 6.** ***DFD Level 1***

Figure 6 separates account access, submission screening, counselor cases and sessions, notifications, and Principal reporting. Asterisks mark repeated entities or data stores. D1 groups persistent database and file records.

Processes read and update permitted records; notification processing checks eligibility and contacts. Principal receives aggregates. Recorded notification attempts do not confirm delivery.

### 4.3.4 Entity Relationship Diagram (ERD)




PK = primary key FK = foreign key UQ = unique ? = nullable FK
All foreign keys reference users.id. Selected attributes are shown.
Junction dots join shared routes; each branch represents a separate foreign key.

**Figure 7.** ***StudentShield entity relationship diagram***

Figure 7 shows twelve tables and sixteen foreign keys from schema.sql. Crow's Foot markers indicate cardinality and optionality. Dots join shared routes with matching references and cardinalities; each branch remains a separate relationship to users.id.

Selected attributes include LRN, registration status, access control, session version, and audit IP metadata. Application checks enforce roles. Principal uses role 3 and admins contact records, without an additional table.

### 4.3.5 Data Dictionary

Tables 5–16 describe the twelve tables in schema.sql. PK means primary key, and FK means foreign key. Nullability follows explicit SQL declarations; implicit TIMESTAMP defaults depend on database settings. Foreign keys identify linked records, while application checks control student and staff permissions. No student records or private contacts are included.

**Table 6.** ***Data dictionary: users***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| name | VARCHAR(255) NOT NULL |
| password_hash | VARCHAR(255) NOT NULL |
| role | TINYINT(1) DEFAULT 0; nullable. 0 Student, 1 Counselor, 2 Super Admin, 3 Principal |
| remember_token | VARCHAR(255) NULL; nullable |
| is_active | TINYINT(1) NOT NULL DEFAULT 0 |
| registration_status | VARCHAR(12) NOT NULL DEFAULT 'approved'; registration decision |
| access_disabled | TINYINT(1) NOT NULL DEFAULT 0; account access restriction |

| Field | Type and constraints |
|---|---|
| auth_version | INT NOT NULL DEFAULT 1; invalidates outdated account sessions |
| is_locked | TINYINT(1) NOT NULL DEFAULT 0 |

**Table 7.** ***Data dictionary: users (continued)***

| Field | Type and constraints |
|---|---|
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |
| updated_at | DATETIME NULL DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP; nullable |

**Table 8.** ***Data dictionary: students***

| Field | Type and constraints |
|---|---|
| user_id | INT PRIMARY KEY; FK → users.id; ON DELETE CASCADE |
| student_id | VARCHAR(50) UNIQUE; nullable |
| lrn | VARCHAR(12) NULL UNIQUE; twelve-digit Learner Reference Number |
| parent_phone | VARCHAR(50); nullable |
| parent_email | VARCHAR(255); nullable |
| section | VARCHAR(100); nullable |
| adviser_name | VARCHAR(255); nullable |
| grade_level | VARCHAR(50) NULL; nullable |

| | |
|---|---|
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |
| updated_at | DATETIME NULL DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP; nullable |

**Table 9.** ***Data dictionary: counselors***

| Field | Type and constraints |
|---|---|
| user_id | INT PRIMARY KEY; FK → users.id; ON DELETE CASCADE |
| counselor_email | VARCHAR(255); nullable |
| counselor_phone | VARCHAR(50); nullable |
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |
| updated_at | DATETIME NULL DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP; nullable |

**Table 10.** ***Data dictionary: admins***

| Field | Type and constraints |
|---|---|
| user_id | INT PRIMARY KEY; FK → users.id; ON DELETE CASCADE |
| email | VARCHAR(255); nullable |
| phone | VARCHAR(50); nullable |
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |

| | |
|---|---|
| updated_at | DATETIME NULL DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP; nullable |

**Table 11.** ***Data dictionary: vent_logs***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| user_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| raw_text | TEXT NOT NULL |
| emotions | VARCHAR(255); nullable |
| risk_percent | DECIMAL(5,2); nullable |
| risk_level | VARCHAR(50); nullable |
| trajectory | VARCHAR(50) DEFAULT 'INSUFFICIENT_DATA'; nullable |
| escalation_probability | DECIMAL(5,4) DEFAULT 0.0000; nullable |
| status | ENUM('Pending', 'Reviewed', 'Flagged') DEFAULT 'Pending'; nullable |
| created_at | TIMESTAMP DEFAULT CURRENT_TIMESTAMP; implicit nullability depends on SQL settings |
| updated_at | DATETIME NULL DEFAULT NULL; nullable |
| deleted_at | DATETIME NULL DEFAULT NULL; nullable |

**Table 12.** ***Data dictionary: bully_reports***

| Field | Type and constraints |
|---|---|

| | |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| reporter_student_id | INT; FK → users.id; ON DELETE SET NULL; nullable |
| reporter_type | ENUM('self', 'bystander') DEFAULT 'self'; nullable |
| victim_student_id | INT NULL; FK → users.id; ON DELETE SET NULL; nullable |
| bully_student_id | INT NULL; FK → users.id; ON DELETE SET NULL; nullable |
| incident_details | TEXT; nullable |
| evidence_file | VARCHAR(255); nullable |
| status | VARCHAR(50) NOT NULL DEFAULT 'pending_review' |
| created_at | TIMESTAMP DEFAULT CURRENT_TIMESTAMP; implicit nullability depends on SQL settings |
| updated_at | DATETIME NULL DEFAULT NULL; nullable |
| deleted_at | DATETIME NULL DEFAULT NULL; nullable |

**Table 13.** ***Data dictionary: audit_logs***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| user_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| action | VARCHAR(50) NOT NULL |
| entity_type | VARCHAR(50) NOT NULL |

| | |
|---|---|
| entity_id | INT NOT NULL |
| ip_address | VARCHAR(45) NULL; nullable |
| user_agent | VARCHAR(255) NULL; nullable |
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |

**Table 14.** ***Data dictionary: sms_logs***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| student_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| parent_number | VARCHAR(50) NOT NULL |
| alert_type | VARCHAR(50) NOT NULL |
| message_content | TEXT NOT NULL |
| sent_status | VARCHAR(50) DEFAULT 'pending'; nullable |
| timestamp | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |

**Table 15.** ***Data dictionary: email_logs***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| student_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |

| | |
|---|---|
| recipient_email | VARCHAR(255) NOT NULL |
| alert_type | VARCHAR(50) NOT NULL |
| subject | VARCHAR(255) NOT NULL |
| message_content | TEXT NOT NULL |
| sent_status | VARCHAR(50) DEFAULT 'pending'; nullable |
| timestamp | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |

**Table 16.** ***Data dictionary: counselor_notes***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| student_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| counselor_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| note_type | ENUM('observation', 'recommendation', 'intervention', 'follow_up', 'general') DEFAULT 'general'; nullable |
| content | TEXT NOT NULL |
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |
| updated_at | DATETIME NULL DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP; nullable |
| deleted_at | DATETIME NULL DEFAULT NULL; nullable |

**Table 17.** ***Data dictionary: case_status_history***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| student_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| status | ENUM('active_monitoring', 'intervention_ongoing', 'parent_consultation_required', 'escalated', 'under_investigation', 'resolved', 'closed') NOT NULL |
| remarks | TEXT NULL; nullable |
| changed_by | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |

**Table 18.** ***Data dictionary: virtual_meetings***

| Field | Type and constraints |
|---|---|
| id | INT AUTO INCREMENT PRIMARY KEY |
| student_id | INT NOT NULL; FK → users.id; ON DELETE CASCADE |
| counselor_id | INT NULL; FK → users.id; ON DELETE CASCADE; nullable |
| meet_link | VARCHAR(255) NOT NULL DEFAULT '' |
| status | VARCHAR(50) NOT NULL DEFAULT 'requested' |
| created_at | DATETIME DEFAULT CURRENT_TIMESTAMP; nullable |

**Software Testing Plan**

Testing will examine individual rules, module communication, and complete student and staff workflows. The matrices below contain planned cases, not completed results. Each run will record its actual output, status, environment, and evidence. Network-dependent cases will use controlled mocks before authorized deployment checks. Synthetic accounts and entries will protect student disclosures.

**Unit Testing**

Existing test files provide a starting point. SecurityConfigTest.php checks CSRF and debug-route settings; AuthenticationSecurityTest.php checks authentication safeguards; GoogleMeetSessionsTest.php covers links and access; and test_risk_history.py checks deterministic history handling. The files do not establish that a test run is complete.

**Table 19.** ***Planned unit test cases***

| Test case | Expected result | Actual result / status |
|---|---|---|
| CSRF configuration | Protected routes keep CSRF controls; AJAX requests follow the configured token rules. | |
| Risk history window | Ignore invalid entries and those older than 14 days; labeled sequences return the expected trajectory. | |
| Crisis safeguards | Reviewed crisis and violent-threat examples preserve local safety rules and expected categories. | |

| | | |
|---|---|---|
| Meet URL validation | Normalize only the accepted Google Meet URL format; reject malformed links. | |

**Integration Testing**

**Table 20.** ***Planned authentication integration tests***

| Test case | Expected result | Actual result / status |
|---|---|---|
| Student access | Active accounts with valid credentials can sign in; invalid, locked, or inactive accounts are rejected. | |
| Staff verification | Counselors, Super Administrators, and Principals complete six-digit email OTP verification unless a valid trusted-device exception applies. | |
| OTP rejection | Incorrect or expired codes do not authorize access. | |
| Role permissions | Students cannot access staff routes; Counselors cannot access Super Administrator user-management routes. | |

**Table 21.** ***Planned vent integration tests***

| Test case | Expected result | Actual result / status |
|---|---|---|
| Save and read vent | Normal storage encrypts text with AES-256-CTR, and authorized staff can read it. Check error handling separately. | |

| | | |
|---|---|---|
| Screening request | Authenticated requests include current text and 14-day risk history; risk and trajectory fields are handled correctly. | |
| ML unavailable | Local keyword safety checks continue when the remote service fails. | |
| Alert eligibility | HIGH triggers eligible counselor triage; CRITICAL may also attempt parent alerts, subject to privacy and notification rules. | |
| Parent opt-out | Counselor-only selection suppresses automated parent alerts for that entry. | |
| Cooldown | A second vent inside the configured 30-second interval is rejected. | |

**System Testing**

**Table 22.** ***Planned incident-report system tests***

| Test case | Expected result | Actual result / status |
|---|---|---|
| Submit report | Self and bystander reports check required fields and keep submitted evidence within the authorized workflow. | |
| Review and verify | Staff review precedes verified-incident parent alert attempts; rejected reports do not trigger verified-incident alerts. | |
| Reject and undo | Rejection, false-report strike handling, and undo follow the implemented status rules. | |
| Cooldown | A second report inside the configured 120-second interval is rejected. | |

**Table 23.** ***Planned dashboard, case, and session tests***

| Test case | Expected result | Actual result / status |
|---|---|---|
| Feed and filters | Search, status/date filters, charts, and refresh reflect matching records. | |
| Record actions | Resolve, archive, restore, and supported bulk actions update the intended records. | |
| Student profile | Selecting a student opens that student's profile and chronological history. The case summary keeps the same student identity. | |
| Google Meet session | Staff publish a new link; assigned participants can join an active session. Unassigned, ended, expired, or reused links are rejected. | |

**Table 24.** ***Planned administration and audit tests***

| Test case | Expected result | Actual result / status |
|---|---|---|
| User management | Super Administrators can create and edit permitted accounts; safeguards prevent self-demotion and self-deletion. | |
| Registration review | Authorized staff can approve or reject registrations while other roles are blocked. | |
| Audit trail | Relevant actions record actor, target, timestamp, and available request metadata. | |
| Notification history | Available attempt logs display correctly without presenting attempts as confirmed delivery. | |

**User Acceptance Testing (UAT) and Evaluation**

The proposed evaluation will use guided task observations, questionnaires, and staff interviews to assess StudentShield. The target sample is 50 evaluators: 10 IT professionals, 25 IT students, 10 non-IT professionals, and 5 client users from Saint Vincent Blessed School of Manila (SVBSM). These are recruitment targets; actual participation and findings will be reported after data collection.

**Data Gathering Technique**

Participants will complete assigned tasks using demonstration accounts and fictional records. Students will test student workflows; staff-role evaluators will test only the functions permitted for their assigned accounts. Researchers will record task completion, errors, completion times, and comments. Two Google Forms questionnaires will be used: User Evaluation for the 40 IT-student, non-IT-professional, and SVBSM evaluators, and IT Professional Evaluation for the 10 technical evaluators. Both will present category headings and individual numbered statements with the same five-point agreement scale and a Not observed or not applicable (N/O) option. Each form will include a brief privacy notice and voluntary participation statement without requesting names, LRN, passwords, contact details, or personal mental-health disclosures. School approvals and any required parental permission and student assent will be obtained before participation. Staff interviews will examine how the tested workflows fit school practice.

**Evaluation Criteria**

The evaluation will follow ISO/IEC 25010:2011 and its eight product-quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The draft User Evaluation contains 15 statements on observable functionality, performance, usability, reliability, and access controls. The draft IT Professional Evaluation contains 24 statements across all eight characteristics. Technical ratings will require relevant source code, documentation, measurements, or controlled test evidence; a demonstration alone will not establish encryption, resource efficiency, or maintainability. Participants will select N/O when evidence is unavailable. The questionnaires will undergo adviser and expert review and pilot testing before distribution. Ratings will supplement task observations and technical tests, not establish ISO certification or clinical validity.

**Sampling Technique**

Purposive, non-probability sampling will recruit 10 IT professionals (20%), 25 IT students (50%), 10 non-IT professionals (20%), and 5 SVBSM client users (10%), for a target of 50 distinct respondents. IT professionals will be selected for relevant professional experience and access to the evidence needed for technical review. The other groups will evaluate assigned demonstration tasks. The five SVBSM participants will provide feedback from intended school users; external evaluators will not be treated as substitutes for client users. Respondent group and tested system role will be recorded separately, and each participant will be counted in only one group. Actual client-role

coverage, recruitment shortfalls, approvals, and the evaluation schedule will be documented. The small purposive client sample will limit generalization to the wider school population.

**Statistical Treatment of Data**

Responses will be summarized by frequency and percentage, $P = (f / n) \times 100$, using the valid response count for each analysis. Item means will use $WM = \Sigma fx / n$, where x is the score from 1 to 5, f is its frequency, and n is the valid numerical response count. Blank and N/O answers will be excluded from numerical calculations and reported separately. User and technical results will remain separate; user results will identify respondent groups and tested roles. No pooled overall mean will combine the two instruments. Actual counts will replace recruitment targets in reporting. Comments and interviews will be summarized by recurring issue without identifying participants.

**Table 25.** ***Proposed five-point agreement scale***

| Score | Agreement response |
|---|---|
| 5 | Strongly agree |
| 4 | Agree |
| 3 | Neither agree nor disagree |
| 2 | Disagree |
| 1 | Strongly disagree |

The five-point scale measures agreement with each questionnaire item. Its labels do not establish system quality or ISO certification. The final approved instrument will define grouped mean-score interpretations; they will not be inferred from unfinished responses.

**Table 26.** ***Planned evaluation measures***

| **Measure** | **Planned calculation / interpretation** |
|---|---|
| Response variation | Sample SD = $\sqrt{[\Sigma f(x - \bar{x})^2 / (n - 1)]}$ for $n > 1$. Report variation alongside item means without assuming universal cutoffs. |
| Task completion | Completed assigned tasks / attempted tasks × 100; report errors and completion times separately. |
| Classifier performance | Use held-out labeled examples; report the confusion matrix, class support, precision, recall, F1, and the averaging method. |

For a stated positive class, precision = TP / (TP + FP), recall = TP / (TP + FN), and F1 = 2PR / (P + R). Accuracy is correct predictions divided by all labeled examples. Multiclass reporting will include per-class and macro or weighted scores, identifying undefined ratios. These measures assess classification, not diagnosis or counseling effectiveness. Questionnaire findings, test statuses, and model scores will be added only when evidence is available.

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